Lecture Notes 9
Government Debt and Fiscal Policy

Intermediate Macroeconomics, EC2211

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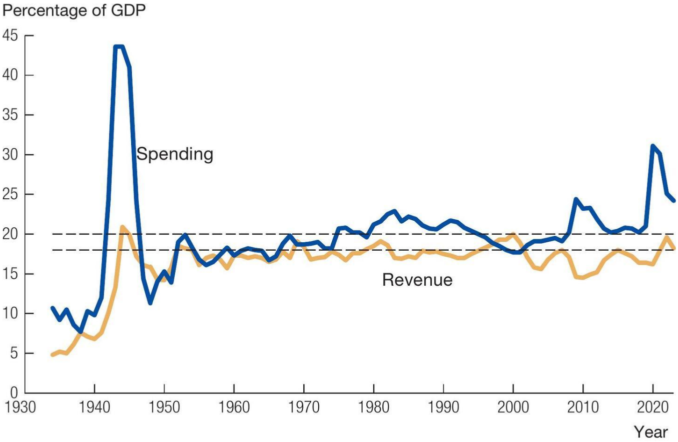
Contents and literature

- The government budget constraint
- Debt dynamics and fiscal sustainability
- The Swedish fiscal framework
- Ricardian equivalence
- Active fiscal policy and automatic stabilizers

Literature:

- Jones (2024), chapters 11 and 18
- Edelberg, Harris and Sheiner (2025), "Assessing the risks and costs of the rising US debt"

U.S. federal government spending and revenue



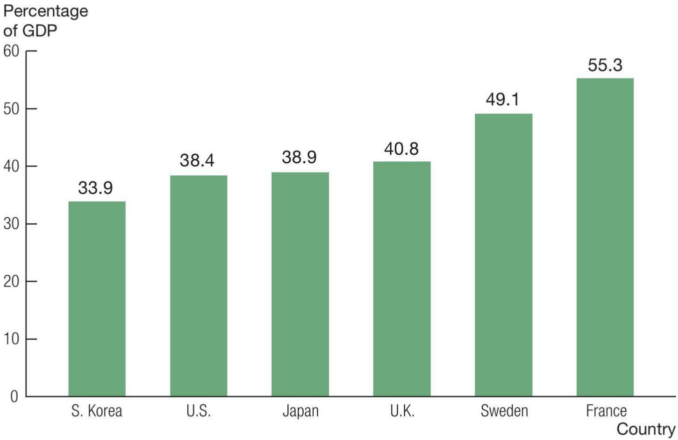
Source: *Economic Report of the President*, 2023, Table B-46.

Figure 18.1 in Jones

U.S. federal debt

Figure 1 in Edelberg et al. (2025)

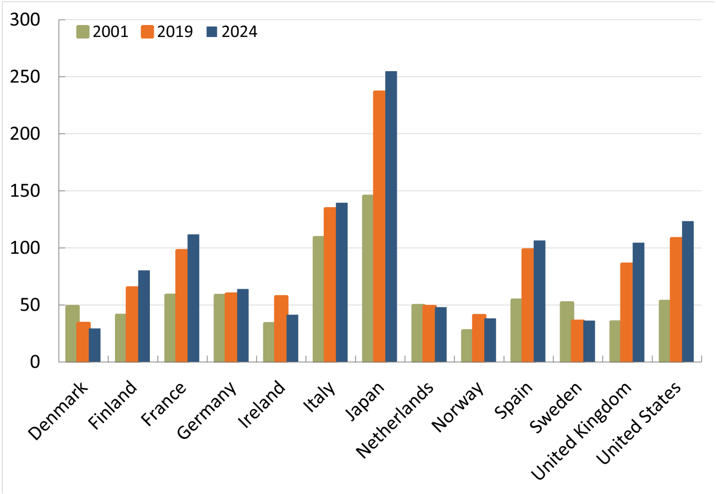
Government spending around the world, 2019



Source: Organisation for Economic Co-operation and Development (OECD) data, data.oecd.org/gga/ general-government-spending.htm. The numbers shown are for 2019 so as not to be distorted by the Covid-19 pandemic.

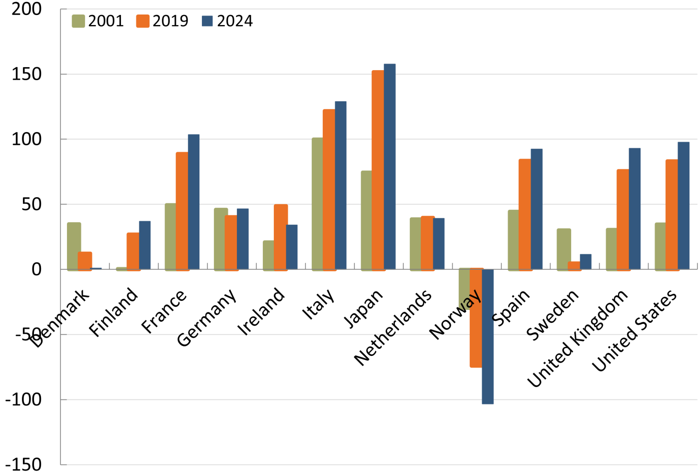
Figure 18.3 in Jones

Gross debt ratios around the world



Gross general government debt, percent of GDP. Source: IMF WEO.

Net debt ratios around the world



Net general government debt, percent of GDP. Source: IMF WEO.

Some questions

- Should we worry about rising debt ratios?
- Can a country run a budget deficit forever?
- When is a country "bankrupt"?

The government budget constraint

The government budget constraint

Notation:

- B_t: government debt ("bonds issued")
- Y_t: nominal GDP
- g: the nominal GDP growth rate
- i_t: the nominal interest rate
- T_t: tax revenue, net of transfer payments
- G_t: government expenditure (excluding interest and transfer payments)
- S_t: the total (fiscal) surplus (government "savings", "net lending")
- S_t^P: the primary (fiscal) surplus

The government's per-period budget constraint

The government budget constraint is

$$G_t + (1 + i_t)B_t = T_t + B_{t+1} \tag{1}$$

The left-hand side in (1) is total expenditure: government expenditure plus repayment of the debt with interest

The right-hand side in (1) is the funding: tax revenue and new debt issued

Jones refers to this per-period budget constraint as the "flow budget constraint"

The government's intertemporal budget constraint

Suppose that there are only two time periods, $t = 1$ and $t = 2$:

- In period 2, no one would lend to the government since there is no future period for repayment. Therefore $B_3 \leq 0$.
- It is not a good idea for the government to have positive savings at the end of the final period. Therefore $B_3 \geq 0$.
- We conclude that $B_3 = 0$

By using $B_3 = 0$ we can combine and reorganize (1) for periods $t = 1$ and $t = 2$ to get:

$$T_1 + \frac{T_2}{1 + i_2} = G_1 + \frac{G_2}{1 + i_2} + (1 + i_1)B_1 \tag{2}$$

This *intertemporal budget constraint* shows that the present value of tax revenue (the left-hand side) must equal the present value of government expenditure including repayment of the initial debt (the right-hand side)

The government's intertemporal budget constraint

What if there are many periods and no final period? When do we have to repay the debt?

Add one more period to the previous example (assume constant interest rate to simplify)

We get:

$$T_1 + \frac{T_2}{1+i} + \frac{T_3}{(1+i)^2} = G_1 + \frac{G_2}{1+i} + \frac{G_3}{(1+i)^2} + (1+i)B_1 \tag{3}$$

The government's intertemporal budget constraint

If we continue like this and add infinitely many periods, we get

$$\sum_{t=1}^{\infty} \frac{T_t}{(1+i)^{t-1}} = \sum_{t=1}^{\infty} \frac{G_t}{(1+i)^{t-1}} + (1+i)B_1 \tag{4}$$

which again just says that the present value of tax revenue has to equal the present value of government expenditure and repayment of debt

The government's intertemporal budget constraint

But what happened to the constraint that $B = 0$ in the final period?

The requirement behind (4) is just that the debt is repaid in present-value terms:

- This would, for example, be the case if debt is held constant over time but the interest rate is positive
- The debt can even grow over time, as long as its growth rate is lower than the interest rate

We will now explore some of this in more detail

Debt dynamics and fiscal sustainability

Government debt dynamics

How does government debt evolve over time?

Recall the government's per-period budget constraint (1):

$$G_t + (1+i)B_t = T_t + B_{t+1} \tag{5}$$

The **primary surplus** is defined as the governments tax revenue minus its expenditure, excluding interest payments:

$$S_t^P = T_t - G_t$$

while the **total surplus** is defined as the governments tax revenue minus its expenditure, including interest payments:

$$S_t = T_t - G_t - iB_t$$

We see that $S_t = S_t^P - iB_t$

Government debt dynamics

Use the primary balance to rewrite the budget constraint (5) as:

$$B_{t+1} = (1+i)B_t - S_t^P \tag{6}$$

Divide by Y_t to express everything in relation to GDP:

$$\frac{B_{t+1}}{Y_t} = (1+i)\frac{B_t}{Y_t} - \frac{S_t^P}{Y_t} \tag{7}$$

Government debt dynamics

Note that

$$\frac{B_{t+1}}{Y_t} = \frac{B_{t+1}}{Y_t}\frac{Y_{t+1}}{Y_{t+1}} = \frac{Y_{t+1}}{Y_t}\frac{B_{t+1}}{Y_{t+1}}$$

Let small letters denote ratios to GDP ($b = B/Y$, $s^P = S^P/Y$ etc) and recall that g denotes the growth rate of GDP

We can then rewrite (7) as

$$(1+g)b_{t+1} = (1+i)b_t - s_t^P \tag{8}$$

Government debt dynamics

We want to understand how the debt ratio, b, develops over time. Subtract b_t on both sides of (8) to get:

$$b_{t+1} - b_t = ib_t - gb_{t+1} - s_t^P \tag{9}$$

or alternatively, in terms of the total surplus:

$$b_{t+1} - b_t = -gb_{t+1} - s_t \tag{10}$$

Stabilizing the debt ratio (based on the primary surplus)

Suppose we want to hold the debt ratio stable over time. Use $b_{t+1} = b_t = b$ in equation (9) to get

$$s^P = (i - g)b \tag{11}$$

Example:

- Suppose that the debt ratio is $b = 50\%$, the nominal interest rate on government debt is $i = 4\%$ and the growth rate of nominal GDP is $g = 3\%$
- The country must then have a primary budget surplus, $(T - G)/Y$, that is one percent of 50 percent, i.e. a surplus of 0.5 percent of GDP, to stabilize its debt ratio

Stabilizing the debt ratio (based on the total surplus)

Suppose we want to hold the debt ratio stable over time. Use $b_{t+1} = b_t = b$ in equation (10) to get

$$s = -gb \tag{12}$$

Example:

- Suppose that the debt ratio is $b = 50\%$ and the growth rate of nominal GDP is $g = 3\%$
- The debt ratio will then remain stable if the country runs a budget *deficit* $s = -gb = -1.5$ percent of GDP
- Note that this is the same example as on the previous slide (if the nominal interest is the same):
  - The primary surplus is 0.5 percent of GDP
  - Interest payments are 2 percent of GDP

So can a country run a deficit forever?

Suppose that $i > g$

The analysis on the previous slides imply that:

- If the debt ratio is positive, the debt ratio will increase (and become infinitely large) if there is a constant primary deficit $s_t^P = s^P \leq 0$
  - If, for example, $s^P = 0$, then $b_{t+1}/b_t = (1+i)/(1+g) > 1$
- It is however possible to run a (small) negative primary deficit forever if the debt ratio is positive
  - Think of Norway, the net-of-growth return $i - g$ on its oil fund allows for a constant wealth-to-GDP ratio even if there are primary deficits
- As long as $g > 0$, the debt ratio will eventually stabilize at some level even if there are permanent total deficits

What if $g \geq i$?

If the growth rate in the economy is higher than the interest rate, the debt ratio can be stabilized at an arbitrary level without the help of a primary surplus!

- If, for example, $i = g$, then $s_t^P = 0$ suffices to hold b_{t+1} equal to b_t even if the debt is very high

But this statement assumes that the interest rate is exogenous, not affected by the debt level

French fiscal woes put cost of sovereign borrowing on a par with corporate debt

Financial Times, September 15, 2025

Vicious circles, doom loops

Recall that equation (11) demonstrates that $s^P = (i - g)b$ is required to stabilize the debt ratio

- It is more difficult to stabilize the debt if the interest rate is high, if GDP growth is low or the debt ratio already is high
- And these things can reinforce one another:
  - If the interest rate goes up, growth typically slows down (IS curve)
  - With lower growth, risk premia on government bonds may increase, i.e. interest rates may increase
  - Fiscal consolidation (to get a higher primary surplus s^P) will reduce demand (shift IS curve in) further

Debt and inflation

"Inflation is always and everywhere a monetary phenomenon"

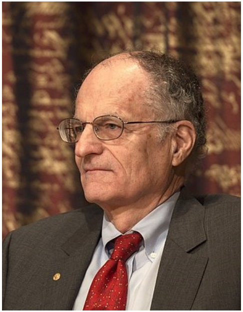
-Milton Friedman

"Inflation is always and everywhere a fiscal phenomenon"

-Thomas Sargent

Thomas Sargent

Nobel prize 2011 (with Christopher Sims) "for their empirical research on cause and effect in the macroeconomy"



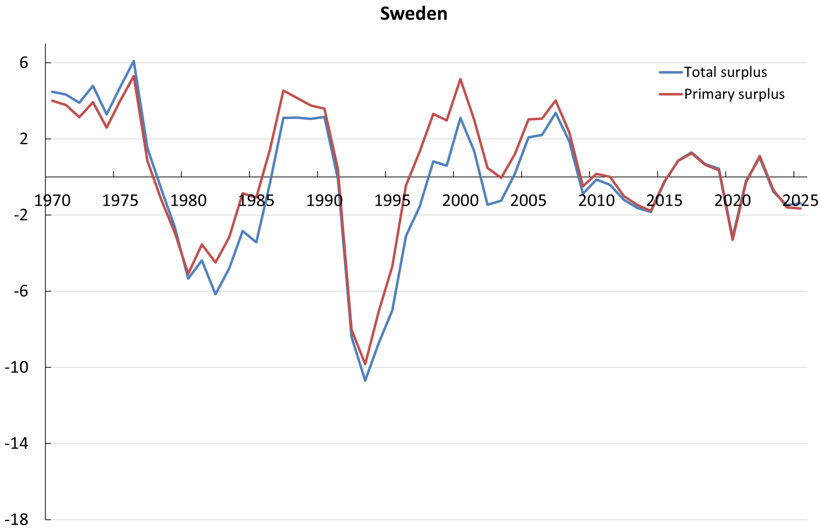
Thomas Sargent
1943-

Debt and inflation

The primary surplus cannot be higher than 100 percent of GDP In practice, it cannot even be close to that level

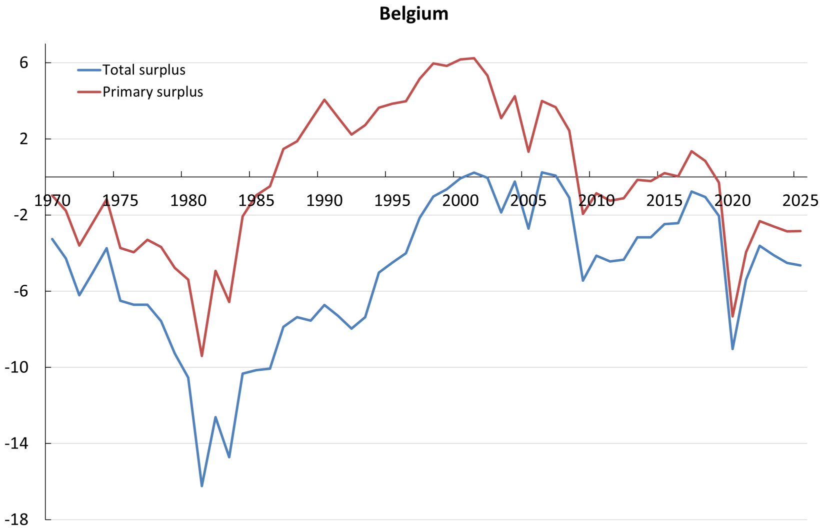
- Economic limits:
  - And recall Laffer curve: output will be negatively affected if tax rate is high
- Political limits:
  - High tax rates with little government spending would not be accepted
- The upper limit on primary surpluses will typically be low
  - Unusual that primary surpluses exceed 5 percent of GDP
  - Episodes of fiscal consolidation after debt crises: Belgium and Sweden in 1990s, Greece in 2010s

Fiscal position



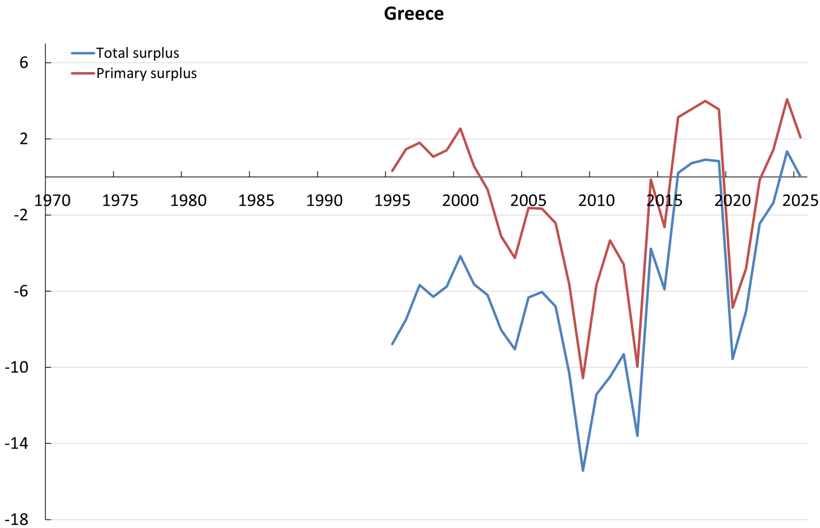
Percent of GDP. Source: OECD EO

Fiscal position



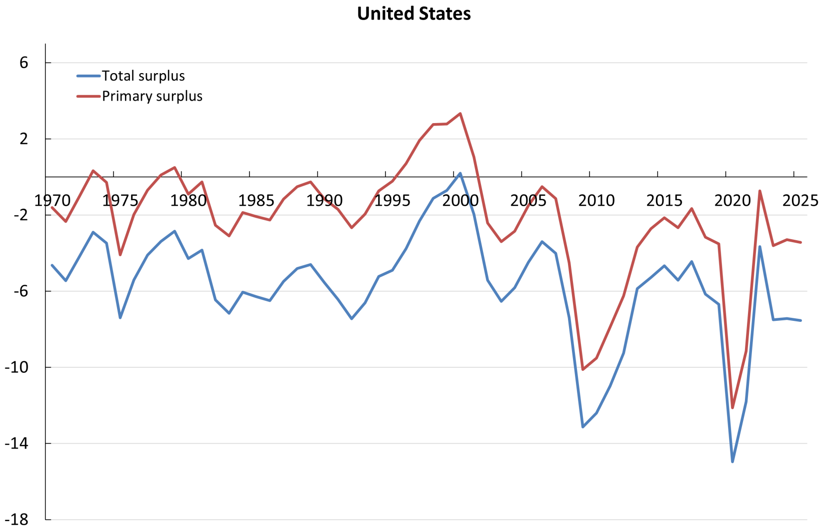
Percent of GDP. Source: OECD EO

Fiscal position



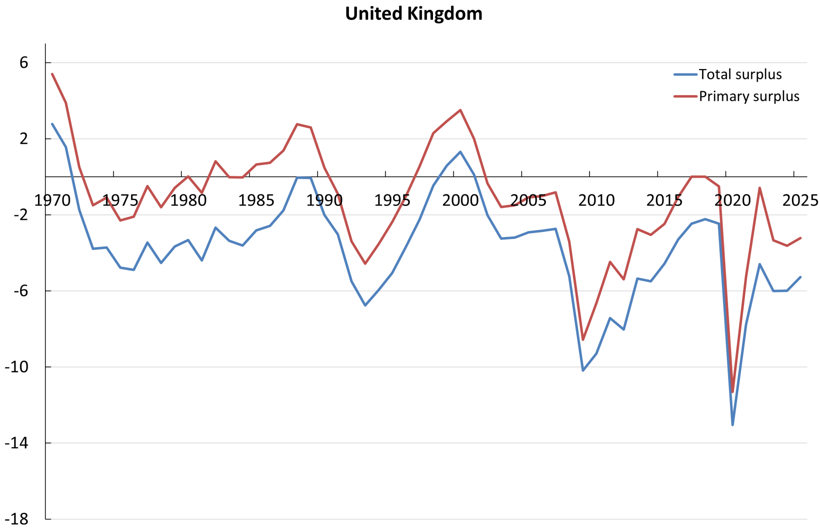
Percent of GDP. Source: OECD EO

Fiscal position



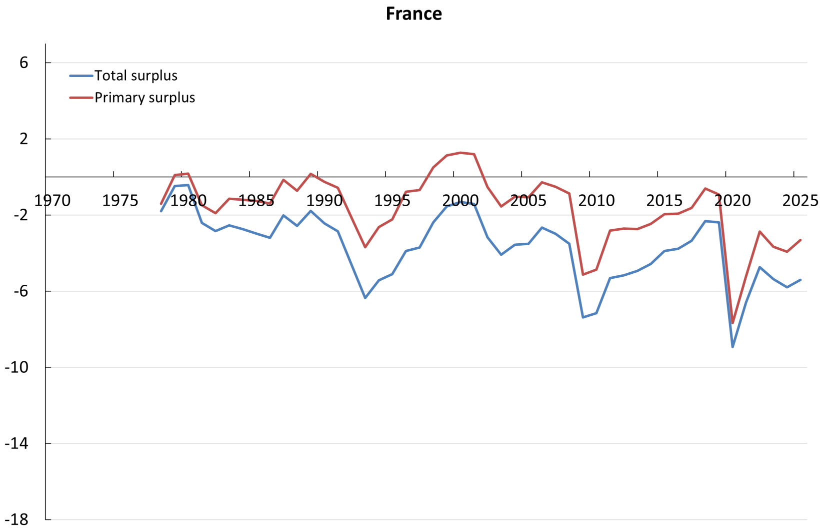
Percent of GDP. Source: OECD EO

Fiscal position



Percent of GDP. Source: OECD EO

Fiscal position



Percent of GDP. Source: OECD EO

Debt and inflation

What if the surplus cannot be raised to levels needed to stabilize the debt?

- Default is one option
- But debt contracts are typically *nominal*
  - The real value of the debt is B/P where P is the price level
- Therefore, other options include:
  - Central bank credits the government's account in the Riksbank balance sheet ("prints money") and gives to the government
  - Central bank promises to purchase government bonds at very low interest rate forever
  - Milder version: central bank pursues more expansionary monetary policy to hold all interest rates low
  - Extreme version: price level increases before central bank has taken action, based on expected fiscal problems

From rate peg to the Treasury-Fed Accord

- Under WWII, the Fed committed itself to maintaining interest rates on public debt low
- The rate peg was maintained in the years after WWII
- 1950-51: Facing rising inflation, Fed repeatedly challenged the rate peg
- March 1951:

  Government debt management separated from monetary policy

  → Fed independence

  → Active market for government bonds

The Treasury-Fed Accord

March 1951

In March 1951, the US Treasury and the Federal Reserve reached an agreement to separate government debt management from monetary policy, laying the foundation for the modern Fed.

William McChesney Martin Jr. is sworn in as Chairman of the Federal Reserve Board of Governors (Photo: Associated Press)

See federalreservehistory.org for further reading

Debt and inflation

- All this demonstrates the importance of fiscal discipline for monetary policy
  - ... and the importance of independent central banks
- Fiscal sustainability is a prerequisite for effective monetary policy with low and stable inflation
  - The European *Stability and Growth Pact* aims at guaranteeing such discipline
- "Monetary financing" is explicitly forbidden in the European treaty
  - Central banks cannot "give" money to governments
  - Central banks cannot purchase assets directly from the government

Independence of the Federal Reserve

POLITICS

Trump says he'll have a Fed 'majority' soon to push rates lower after firing Cook

PUBLISHED TUE, AUG 26 2025-2:54 PM EDT | UPDATED TUE, AUG 26 2025-3:38 PM EDT

Finance & economics | Cook out?

Trump "fires" Lisa Cook, escalating his war on the Federal Reserve

There is little precedent: no Fed governor has been dismissed for cause before

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Trump Has Draft of Letter to Fire Fed Chair. He Asked Republicans if He Should Send It.

Trump calls for 'bigger' interest rate cut ahead of Fed meeting

By Reuters

September 15, 2025 2:52 PM GMT+2 · Updated September 15, 2025

Supreme Court decision on Donald Trump's firing of Lisa Cook looms over the Fed

President could purge other top officials if justices allow him to remove the central banker

TRUTH.

Truth Details

1856 replies

Donald J. Trump

@realDonaldTrump

"Too Late" MUST CUT INTEREST RATES, NOW, AND BIGGER THAN HE HAD IN MIND. HOUSING WILL SOAR!!! President DJT

7.24k ReTruths 35.3k Likes 9/15/25, 2:39 PM

The Swedish fiscal framework

The Swedish fiscal-policy framework

- A prominent role for the total fiscal balance, S_t
- Initial target was $s = S/Y = 2\%$
- Current target is $s = 0.33\%$
- Agreement to reduce the target to $s = 0$ from 2027
- Currently several exemptions

The Swedish fiscal-policy framework

Note that (5) implies that

$$\Delta B_{t+1} = G_t - T_t + iB_t = -S_t \tag{13}$$

In other words,

- The debt is constant if $S = 0$
  - If the economy is growing ($g > 0$) the debt ratio (B/Y) shrinks towards zero
- The debt shrinks over time if $S > 0$, and it eventually becomes negative
- More generally, equation (12) shows that the debt ratio will stabilize at

$$b = -\frac{s}{g} \tag{14}$$

- A surplus target thus means that the debt ratio will settle down at a negative level
  - Example, with $s = 0.33$ and nominal GDP growth at 3 percent per year
  - ... the steady-state debt ratio would be $b = -0.11$, i.e. a net wealth of 11 percent of GDP
- With $s = 0$, the debt ratio will stabilize at 0
  - Why would b shrink to zero?
  - With $s = 0$, there is a primary surplus that precisely covers interest on the debt every period
  - The debt, B is then *constant*, but the economy grows so B/Y shrinks

Ricardian equivalence

Ricardian equivalence

Recall the household's consumption/savings choice in Lecture Notes 5b

The household's budget constraint is:

$$C_1 + \frac{C_2}{1+i} = Y_1 - T_1 + \frac{Y_2 - T_2}{1+i} \tag{15}$$

where $Y - T$ is the household's income after tax

In a two-period setting and assuming that there is no initial public debt ($B_1 = 0$), the government's budget constraint (2) is:

$$G_1 + \frac{G_2}{1+i} = T_1 + \frac{T_2}{1+i} \tag{16}$$

Ricardian equivalence

Use (16) to substitute for taxation in the household's budget constraint (15) to see that

$$C_1 + \frac{C_2}{1+i} = Y_1 + \frac{Y_2}{1+i} - \left(G_1 + \frac{G_2}{1+i}\right) \tag{17}$$

According to this analysis:

- The representative households will have to pay for the government's expenditure
- The timing of taxes "should" not affect the household's behavior
- The household consumption decision is affected by the present value of income and the present value of the government's expenditure

Assumptions behind Ricardian equivalence

1. Forward-looking households
2. Households understand the intertemporal government budget constraint
3. Lower taxes today do not imply lower future government consumption
4. Households are not credit constrained
5. The current generation cares about future generations

The effects of a tax cut

Consider a tax cut by ΔT_1 in period 1

A period-1 tax cut ΔT_1 must be met by a period-2 tax increase of magnitude $(1+i)\Delta T_1$

Present value of the future tax increase:

$$PV = \frac{(1+i)\Delta T_1}{1+i} = \Delta T_1 \tag{18}$$

The tax cut has no effect on the lifetime income of households and therefore no effect on their consumption

A temporary increase in government expenditure

- Consider now a temporary increase in government expenditure, $\Delta G_1 > 0$, and suppose there are many time periods
- Suppose that households anticipate a future tax increase to pay for the higher G
- The future tax increase lowers lifetime income, causing households to consume less, but because of consumption smoothing, the decrease in consumption in any given period is small
- Since $\Delta G_1 > 0$ and $\Delta C_1 < 0$ but the former dominates the latter, a temporary increase in government expenditure will increase aggregate demand in the current period

A permanent increase in government expenditure

- Next, consider a *permanent* increase in government expenditure
- Households now anticipate a *permanent* future tax increase to pay for the higher G
- The permanent future tax increase lowers lifetime income by more than under a temporary increase in G, causing private consumption to decrease more
- Since $|\Delta C_1|$ is much larger in this case, it will offset $\Delta G_1 > 0$, so that a permanent increase in G will increase aggregate demand very little (if at all)

Active fiscal policy and automatic stabilizers

The objective of fiscal policy

- So far today we talked about fiscal sustainability: whenever the government implements a fiscal policy measure it needs to be sustainable in the long run
- Fiscal policy comprises a great range of measures that the government can implement to meet certain objectives
- Example objectives of fiscal policy:
  - Contribute to an efficient allocation of resources
  - Redistribution
  - Meet objectives related to climate change
  - Stabilise the business cycle (our focus for the remainder of the lecture)

Two types of fiscal policy

1. Automatic stabilizers: automatic changes in tax revenue and government expenditure occurring over the business cycle

2. Active (discretionary) fiscal policy: active decisions on tax rates and government spending with the ambition to mitigate economic fluctuations

Automatic stabilizers

- Automatic changes in the fiscal balance due to changes in tax revenue and government expenditure occurring over the business cycle
- Mitigate economic fluctuations without requiring active decisions
- Examples: income taxes, unemployment insurance, some transfers
    - In a boom, both labor and capital income increase: With fixed tax rates, tax revenue increase without political decisions ($T - G$ then increases)
    - In a boom, unemployment falls: For a given unemployment insurance system, transfer payments fall ($T - G$ then increases)

Active (discretionary) fiscal policy

- Active decisions on tax rates and government spending
- Examples on active fiscal policy measures in a recession:
  - Decision to reduce tax rates
  - Decision to raise replacement rate in the unemployment insurance system
  - Send checks to all households
  - Subsidize firms' investments (typically through tax deductions)
  - Raise public expenditure, e.g. through large public investment projects
  - ...
- What are the effects on the economy?
  - Recall IS curve: Higher G shifts the IS curve out, subsidized investment also shifts IS curve out
  - If household consumption responds to *disposable* income, a multiplier effect is likely, and tax cuts then stimulate the economy

Active fiscal policy vs automatic stabilizers

- Many economists are sceptical against active fiscal policy:
  - Estimates of the multiplier are often modest (but difficult to estimate, and estimates vary!)
  - Politically easy to stimulate in recession, more difficult to consolidate in boom
  - "Lags": it takes time to formulate these policies, to get them approved in parliament and then to implement policies and for them to have effect
- But in practice, active fiscal policy is often implemented in recessions
- And the view on active fiscal policy has become a bit more positive in the last decades:
  - Large shocks: GFC and pandemic
  - Binding lower bound on policy rates
  - "Monetary policy cannot be the only game in town"

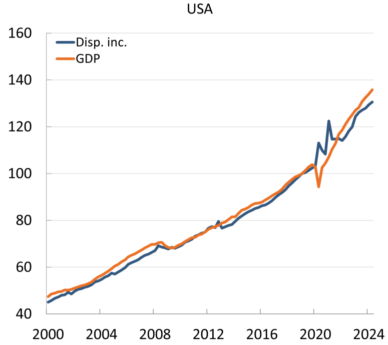
Active fiscal policy vs monetary policy

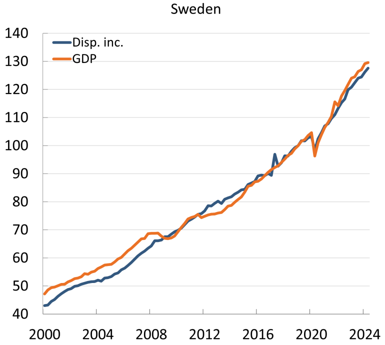
- Note that both fiscal and monetary policy can be used to stabilize the economy after shock
  - Who is responsible for taking action?
  - Do fiscal and monetary policies need to be coordinated?
- A special case: fiscal policy more important when the interest rate is at the effective lower bound
- If the interest rate is above the lower bound, more expansionary fiscal policy may be met by more contractionary monetary policy ("crowding out")...

2020/2021 fiscal support in the United States

- March/April 2020
  - Immediate fiscal impulse corresponding to 9 percent of GDP and an additional 5 percent of GDP in tax deferrals and guarantee programs (source: Bruegel, 2020)
  - Example: $1,200 per adult, checks
- Fall 2020 / Spring 2021
  - Expanded unemployment benefit: additional $300 *per week* in unemployment benefit
  - New checks, $1,400 per adult
  - Tax credits
  - ...

Disposable income increased in the United States during and after covid

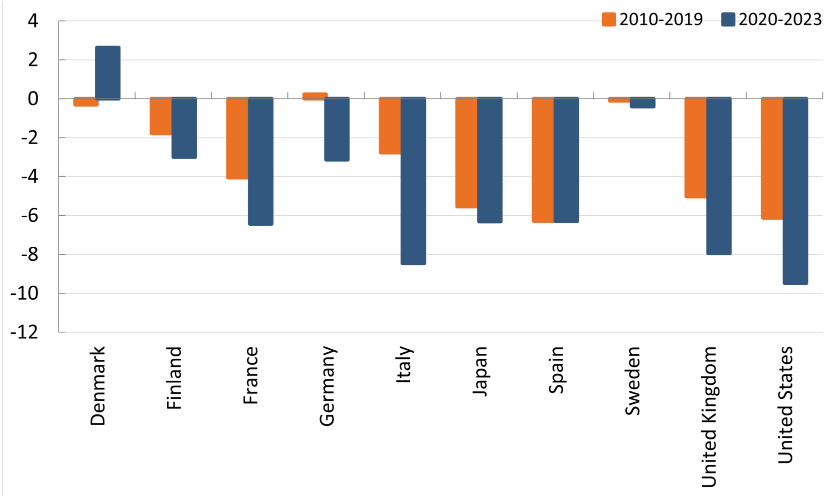




How to think about the covid recession in the IS-MP-PC and AS-AD frameworks?

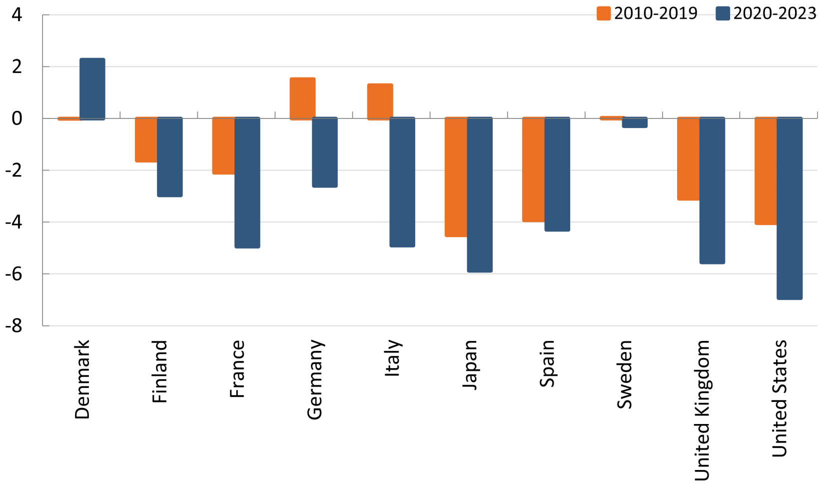
- A fall in demand or supply?
- Arguably, much was initially driven by *supply*
    - Economies closed
    - Global value chains, international trade, disrupted
- Would/could have generated sharp fall in demand without support measures
- Was magnitude and duration of support measures warranted?

Total surplus around the globe



General government net lending, percent of GDP. Source: IMF WEO.

Primary surplus around the globe



General government primary net lending, percent of GDP. Source: IMF WEO.